Experiment : 4
CNN-Based Digit Classification
Handwritten Digit Recognition Using a Convolutional Neural Network

Objectives: To develop and evaluate a Convolutional Neural Network (CNN) for classifying handwritten digits from 0 to 9 using the MNIST dataset. The model performance is assessed using test accuracy, training/validation curves, and a confusion matrix.

Procedure:

1. Import the required Python libraries, including TensorFlow/Keras, NumPy, Matplotlib, and scikit-learn.
2. Load the MNIST dataset, containing 60,000 training images and 10,000 test images of handwritten digits.
3. Normalize pixel values to the range 0 to 1 and reshape each grayscale image to 28 × 28 × 1 for CNN input.
4. Build the CNN using convolution and max-pooling layers, followed by a flatten layer, a dense layer, dropout, and a 10-class softmax output.
5. Compile the model with the Adam optimizer and sparse categorical cross-entropy loss, then train for up to 10 epochs using a validation split.
6. Evaluate the trained model on the test dataset, generate predictions, and calculate the confusion matrix and classification metrics.
7. Plot training/validation accuracy and loss, test a handwritten digit image, and save the trained model.

Table column:

Parameter	Specification / Result
Dataset	MNIST: 60,000 training images and 10,000 test images
Input image	28 × 28 pixels, grayscale, normalized to 0–1
Output classes	10 classes representing digits 0–9
CNN architecture	Conv2D (32 filters) → MaxPooling → Conv2D (64 filters) → MaxPooling → Flatten
Classifier layers	Dense (128, ReLU) → Dropout (0.3) → Dense (10, Softmax)
Training configuration	Adam optimizer; sparse categorical cross-entropy; batch size 128; up to 10 epochs
Trainable parameters	225,034
Test accuracy	99.13%

Result: The CNN model achieved an overall test accuracy of 99.13% on the MNIST test set. The confusion matrix shows that most predictions lie on the main diagonal, indicating that the model correctly classifies the majority of handwritten digits. In the sample uploaded-image test, digit 7 was correctly identified as 7.

Result Images

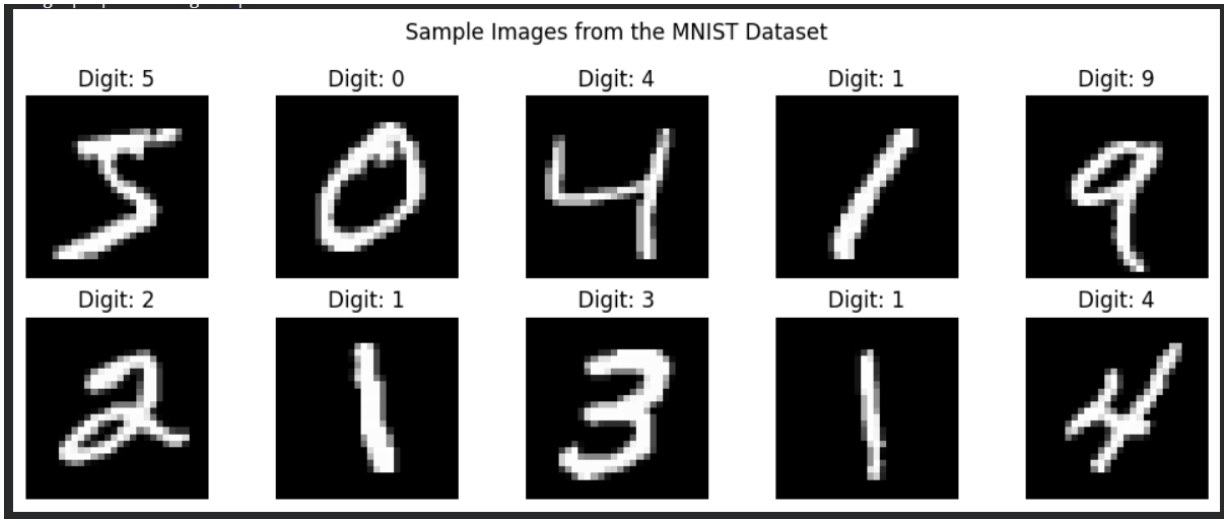


Figure 1. Sample handwritten digit images from the MNIST dataset.

CNN Model Summary:

Model: "sequential"

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 26, 26, 32)	320
max_pooling2d (MaxPooling2D)	(None, 13, 13, 32)	0
conv2d_1 (Conv2D)	(None, 11, 11, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 5, 5, 64)	0
flatten (Flatten)	(None, 1600)	0
dense (Dense)	(None, 128)	204,928
dropout (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 10)	1,290

Total params: 225,034 (879.04 KB)
Trainable params: 225,034 (879.04 KB)
Non-trainable params: 0 (0.00 B)

Figure 2. CNN model summary and trainable parameters.

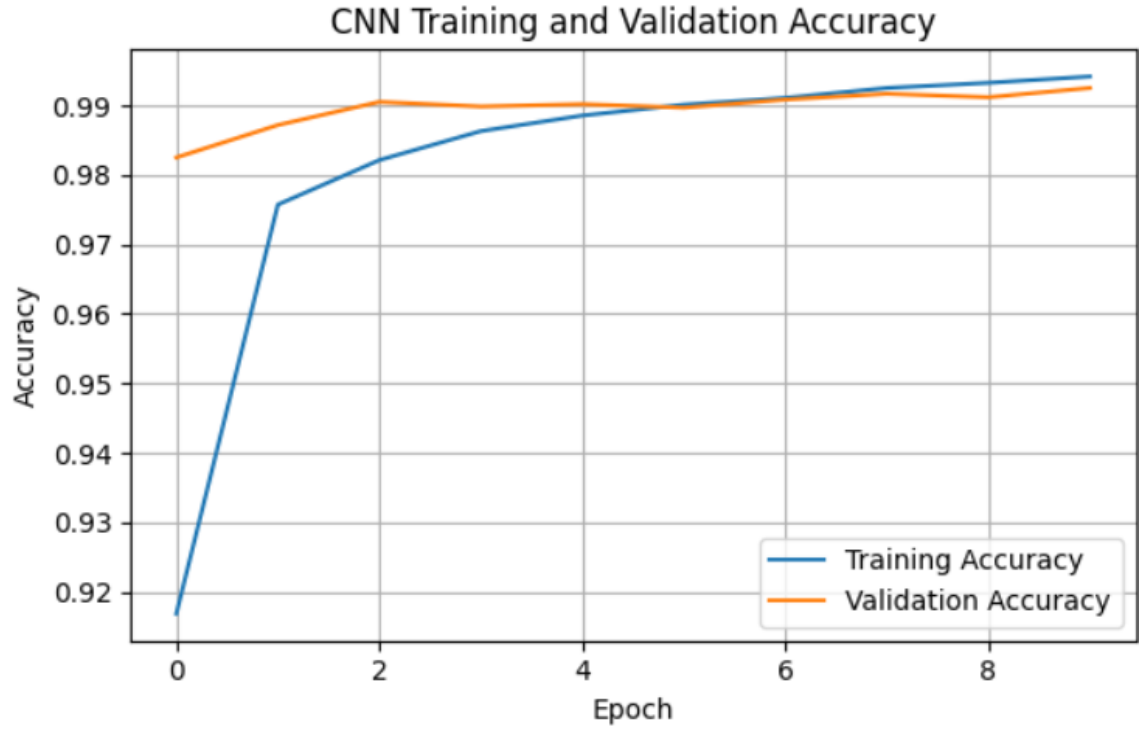


Figure 3. Training and validation accuracy across epochs.

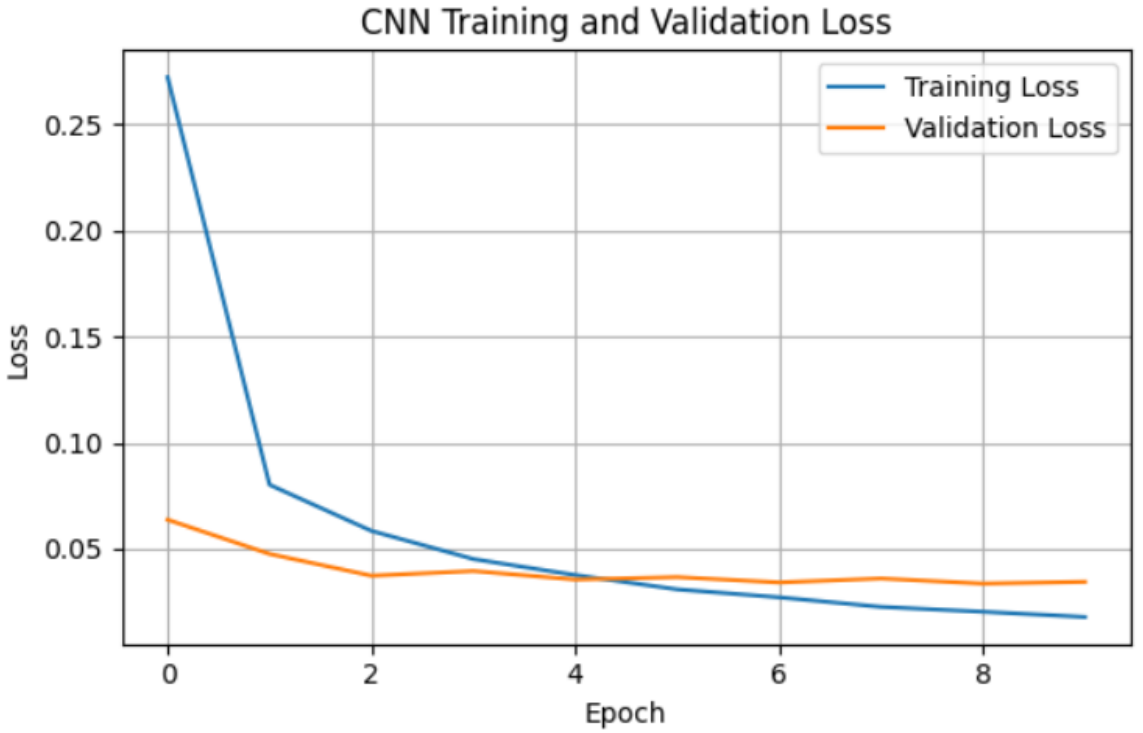


Figure 4. Training and validation loss across epochs.

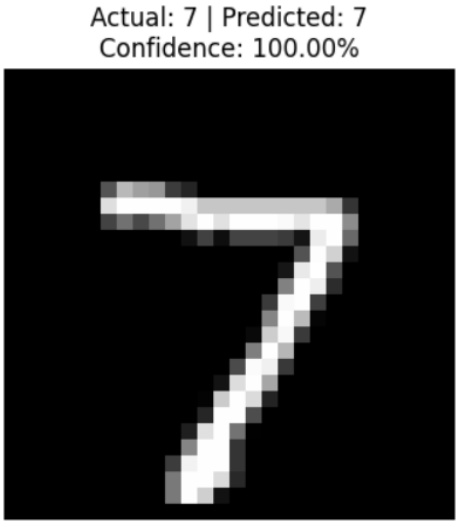


Figure 5. Example prediction: the actual digit 7 was predicted as 7.

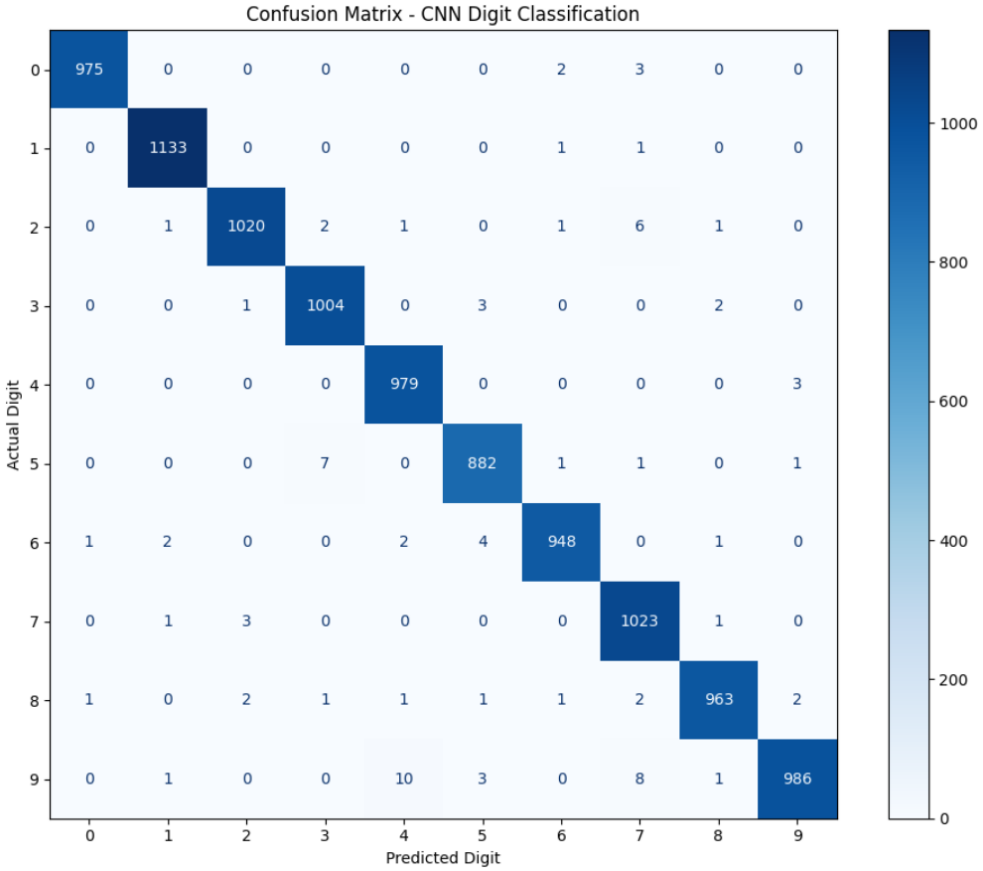


Figure 6. Confusion matrix for CNN-based digit classification.

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Overall Test Accuracy: 99.13%
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Figure 7. Overall test accuracy output.

Table column: Digit-wise classification results

Digit / Total	Test samples	Correct	Misclassified	Recall / accuracy (%)
0	980	975	5	99.49%
1	1,135	1,133	2	99.82%
2	1,032	1,020	12	98.84%
3	1,010	1,004	6	99.41%
4	982	979	3	99.69%
5	892	882	10	98.88%
6	958	948	10	98.96%
7	1,028	1,023	5	99.51%
8	974	963	11	98.87%
9	1,009	986	23	97.72%
Overall	**10,000**	**9,913**	**87**	**99.13%**

Final Result: The CNN-based model successfully classified handwritten digits 0-9 with 99.13% accuracy on the MNIST test set.